GARGANTUA

Quasinormal Mode Frequencies – Averaged Over All Data

DM Family

Mode	Theory		Observed/Theory - 1	
(l,m,n)	Re(ω) M	Im(ω) M	Re(ω) M	Im(ω) M
(2,1,0)	0.6664799	0.05541304	0.000054±23	0.00038±44
(2,1,1)	0.6665907	0.1662391	0.000008±8	0.00025±26
(2,1,2)	0.6667016	0.2770652	0.000040±17	0.00039±39
(2,1,3)	0.6668124	0.3878913	0.000016±24	0.00051±8
(2,1,4)	0.6669232	0.4987174	0.000003±25	0.00005±8
(2,0,0)	0.5235067	0.0809975	0.000057±10	0.00017±19
(2,0,1)	0.5236687	0.2429925	0.000029±9	0.00065±13
(2,0,2)	0.5238307	0.4049875	0.000005±31	0.00042±15
(2,0,3)	0.5239927	0.5669825	0.000023±12	0.00039±50
(2,0,4)	0.5241547	0.7289775	0.000041±61	0.00003±46
(3,2,0)	1.0749379	0.03192427	0.000014±91	0.00009±71
(3,2,1)	1.0750018	0.09577282	0.000019±32	0.00021±24
(3,2,2)	1.0750656	0.1596214	0.000004±25	0.00006±21
(3,2,3)	1.0751295	0.2234699	0.000024±14	0.0011±19
(3,2,4)	1.0751933	0.2873185	0.000032±38	0.00007±28
(3,1,0)	0.8623969	0.06574082	0.000004±74	0.00051±27
(3,1,1)	0.8625284	0.1972225	0.00039±1	0.00016±9
(3,1,2)	0.8626599	0.3287041	0.000019±35	0.00057±41
(3,1,3)	0.8627914	0.4601857	0.000030±35	0.00002±21



Fig. 18.1. The first page of the data that Yang and Zimmermann prepared for Romilly to show to Amelia Brand. *[Prop from* Interstellar*, used courtesy of Warner Bros. Entertainment Inc.]*